

1. What will be the output of the code below?

```
int b = 10, c = -1;  
int a = ++b+c++;  
System.out.print(""+a+b+c);
```

- A. will not compile
- B. 22
- C. 23
- D. 11110
- E. 10110
- F. 10111

2. What will be the output of the code below?

```
int x = 11, z = 2, w = 15;  
float f = 10.5f;  
f += f++;  
var result = z*(x++)-(--w)+f;  
System.out.println(result);
```

- A. 31
- B. 30
- C. 28
- D. 29
- E. 30

3. What will be the output of the code below?

```
String a = "java", b = "j";  
String c = "jdk", d = "";  
b = (b + "ava").intern(); d += "jdk";  
System.out.println("" + (a==b) + (d==c) + d.equals(c));
```

- A. falsefalsetrue
- B. true>true>true
- C. truefalsetrue
- D. falsefalsetrue
- E. none of them

4. What will be the output of the code below?

```
var sb = new StringBuilder("ocp");  
var sb2 = sb.append(" java");  
var sb3 = new StringBuilder("ocp java");  
System.out.println("" + (sb == sb2) + (sb2.equals(sb3)) + (sb2 == sb3));
```

- A. truefalsefalse
- B. truetruefalse
- C. truefalsetrue
- D. truetruetrue
- E. none of them

5. What will be the output of the code below?

```
String block = ""
    block 1"".trim();
var liter = "block";
int one = 1;
System.out.print(("block" + " 1") == block);
System.out.print(("block " + one) == liter);
```

- A. false>true
- B. false>false
- C. true>true
- D. true>false
- E. none of them

6. What is the output of the code below?

```
BigDecimal up = new BigDecimal("2020.4").setScale(0, RoundingMode.HALF_UP);  
BigDecimal even = new BigDecimal("2020.5").setScale(0,  
RoundingMode.HALF_EVEN);  
BigDecimal down = new BigDecimal("2020.6").setScale(0,  
RoundingMode.HALF_DOWN);  
System.out.print(up + "-" + even + "-" + down);
```

- A. 2020-2020-2021
- B. 2020-2021-2021
- C. 2021-2021-2020
- D. 2021-2020-2021

7. Select statement(s) that will make this code work if we put it in the blank.

```
public void catchException() throws Error {  
    _____  
}
```

- A. throw new Exception()
- B. throw new RuntimeException()
- C. throw new IllegalAccessException()
- D. try {} catch {}
- E. System.out.println("error");

8. What is the output of the code below?

```
int array [] = new int [5];
try {
    System.out.println(array[5]);
} catch (IllegalArgumentException e) {
    System.out.print("error");
} catch (RuntimeException e) {
    System.out.print("runtime");
} catch (Exception e) {
    System.out.print("exception");
} finally {
    System.out.print("finally");
}
```

- A. runtimefinally
- B. finallyruntime
- C. errorfinally
- D. exceptionfinally
- E. finallyexception

9. What is the output of the code below?

```
public void throwOneMore() throw FileNotFoundException {  
    int div = 0;  
    try {  
        int result = 10/div;  
    } catch (RuntimeException e) {  
        System.out.print("div by zero");  
    } catch (Exception e) {  
        System.out.print("exception");  
        throw new FileNotFoundException();  
    } finally {  
        System.out.print("0");  
    }  
}
```

- A. div by zero0
- B. code will not compile
- C. div by zero
- D. exception
- E. exception0

10. Select statements that are true about code below.

```
public class GCollector {  
    public String name;  
  
    public void setName(String name) {  
        this.name = name;  
    }  
    public static void main(String[] args) {  
        String a = new String("a");  
        String b = new String("b");  
        String c = a;  
        new GCollector().setName(b);  
        a = null;  
        b = null;  
        System.out.println(c);  
    }  
}
```

- A. String("a") - is eligible for gc right after a=null line.
- B. String("b") - is eligible for gc right after b=null line.
- C. String("a") and String("b") will live until end of program.
- D. variable c will loose its reference after line a=null.
- E. code will not compile

11. What is the output of the code below?

```
int age = 10;  
var weight = (age++) + 2;  
if(weight > 13) {  
    System.out.print("weight is 13");  
} else if((++age)*weight%2 == 0) {  
    System.out.println("even");  
} else {  
    System.out.println("odd");  
}
```

- A. odd
- B. even
- C. weight is 13
- D. weight is 13even
- E. weight is 13odd
- F. code will not compile

12. What is the output of the code below?

```
var now = LocalDateTime.now();  
now = now.plus(Period.of(1,0,0));  
now = now.plusHours(2);  
System.out.println(now);
```

- A. current time in default timezone + 2 hours and 1 year
- B. current time in UTC-00 + 2 hours and 1 year
- C. code will not compile
- D. will throw UnsupportedOperationException
- E. none of them

13. What is the output of the code below?

```
var startDate = LocalDate.of(2025, 2 , 1);  
var endDate = LocalDate.of(2025, 0 , 1);  
System.out.println("between : " + Period.between(startDate, endDate));
```

- A. between : P-1M
- B. between : P-2M
- C. code will not compile
- D. will throw RuntimeException

14. Select statements that will fix the code below.

```
public enum EvenNumber {  
    TWO, FOUR, SIX, EIGHT  
}  
  
...  
  
var evenNum = EvenNumber.TWO;  
int result = switch (evenNum) {  
    case TWO -> 2; break;  
    case FOUR -> {  
        return 4;  
    }  
    case SIX: {  
        yield 6;  
    }  
};
```

- A. remove yield from case SIX; add return 6;
- B. add case EIGHT
- C. in case FOUR replace keyword return to yield
- D. case SIX change symbol : to arrow ->
- E. without default case it will not work, even if we added case EIGHT
- F. case TWO remove keyword break

16. Select statements that are true about `Period` and `Duration` in java.

- A. `Period` and `Duration` can be used with `LocalDate`, `LocalDateTime` and date classes.
- B. `Period` can be used with `LocalDate`, `LocalDateTime` and `ZonedDateTime`.
- C. `Duration` can work with `LocalDate`.
- D. `Period.of(1,2,3)` will create period with 1 year, 2 months and 3 days
- E. `Duration` works with week, days, hours and seconds

17. Select statements that are true about exceptions in java.

- A. try with resource works with objects that implements Closeable.
- B. try with resources, do not release resources implicitly.
- C. try catch finally block works more effectively with classes that implements AutoClosable rather than try with resource.
- D. finally block do not throw exceptions.
- E. When using finally programmer will not have a resource leak.

18. What is the result of the code below?

```
try (FileReader reader = new FileReader("file.txt");
    var bufferedWriter = new BufferedWriter(reader);
    bufferedWriter = null;) {
    throw new IllegalArgumentException();
} catch (Exception e) {
    System.out.println("Exception");
} catch (RuntimeException e) {
    System.out.println("RuntimeException");
}
```

- A. Exception
- B. RuntimeException
- C. ExceptionRuntimeException
- D. code will not compile
- E. none of the above

19. What is the result of the code below?

```
try (var reader = new FileReader("file.txt")) {  
    throw new FileNotFoundException();  
} catch (Error e) {  
    System.out.print("Error");  
throw new RuntimeException();  
} catch (Exception e) {  
    System.out.print("Exception");  
}
```

- A. ErrorException
- B. Error
- C. Exception
- D. code will not compile
- E. none of the above

20. Select statements that are correct about garbage collection.

```
class GCZoo {  
    public void referenceZoo(GCZoo zoo) {  
        GCZoo localZoo = zoo;  
    }  
    public static void main(String[] args) {  
        GCZoo zooOne = new GCZoo();  
        GCZoo zooTwo = new GCZoo();  
        new GCZoo().referenceZoo(zooOne);  
        zooOne = null;  
        zooTwo = null;  
        System.gc();  
    }  
}
```

- A. object on line 6 is eligible for garbage collection after line 9
- B. GCZoo object on line 8 is not eligible until the end of program
- C. GCZoo object on line 8 is eligible right after execution
- D. object on line 6 has additional reference on line 3 and will live until end of program
- E. object on line 7 is eligible for garbage collection after line 10

21. What is the output of the following code?

```
Object obj = 3.14;
String result = switch (obj) {
    case Integer i -> "Integer: " + i;
    case String s -> "String: " + s;
    case Double d -> "Double: " + d;
    default -> "Unknown";
};
System.out.println(result);
```

- A. Integer: 3.14
- B. Double: 3.14
- C. Unknown
- D. Compilation error

22. What is the output of the code below?

```
public class Main {
    public static void main(String[] args) {
        var a = "aa";
        String b = a + "b", c="aabb";
        Main m = new Main();
        if(b == (a + "b")) {
            m.runSwitch(a);
            m.runSwitch(b);
        } else {
            m.runSwitch(c);
            if(c == new String("aabb").intern()) {
                System.out.println("c==d");
            }
        }
    }
    public void runSwitch(String a) {
        var aa = switch (a) {
            case "aa" -> "equals aa";
            case "aab" -> "equals aab";
            default -> "default";
        };
        System.out.print(aa + " ");
    }
}
```

- A. code will not compile
- B. equals aa equals aab
- C. equals aa
- D. default
- E. default c==d

23. Select correct statement.

- A. String is immutable.
- B. StringBuffer and StringBuilder are mutable.
- C. StringBuilder is a synchronized version of StringBuffer.

- A. A,C
- B. B, C
- C. A, B, C
- D. A, B
- E. A

24. What is the output of the code below?

```
public class Main {  
    public static int a = 0;  
    public static void main(String[] args) {  
        var b = 7;  
        int c = b+++a;  
        var d = --c + (++a) - b;  
        System.out.print(c + "-" + b + "-" + d);  
    }  
}
```

- A. 6-8--1
- B. code will not compile
- C. 7/7/2002
- D. 7/7/2000
- E. 8/7/2001

25. What is the output of the code below?

```
public class Q2 {  
    public static void main(String[] args) {  
        var d = 11; var k = -1;  
        var c += d;  
        double dd = d * 9.0 + c / -1 + (--k);  
        var res = (int)dd % 4;  
        System.out.println(res);  
    }  
}
```

- A. code will not compile
- B. 3
- C. 0
- D. 1
- E. 2

26. Select statement(s) that are true about code below.

```
public class Q3 {  
    public static void main(String[] args) {  
        int a = 11; short b = 9;  
        long c = 8; double f = 3;  
        var r1 = a*b;  
        var r2 = c+r1;  
        var r3 = r2 + f;  
    }  
}
```

- A. r1 is int
- B. r2 is int
- C. r2 is long
- D. r3 is long
- E. r3 is double
- F. r1 is short

27. What is the output of the code below?

```
public class Q30 {  
    public static void main(String[] args) {  
        var cond = false;  
        int a = 7;  
        var b = (byte)(Byte.MAX_VALUE + 1);  
        int k = 2;  
        k--;  
        a = a + b;  
        System.out.println(~k + " | " + a + " | " + (int)cond);  
    }  
}
```

- A. -2 | -121 | false
- B. -2 | 137 | 0
- C. -2 | -121 | 0
- D. -1 | 137 | 0
- E. code will not compile

28. What is the output of the code below?

```
public class MathApi {  
    public static void main(String[] args) {  
  
        var random = Math.random();  
        var ternary = random == 0.0 ? 0.1 : random;  
        var ceil = Math.ceil(ternary);  
        var pow = Math.pow(3, 2) * ++ceil;  
        System.out.print(pow);  
        System.out.print("-");  
        var sqrt = Math.sqrt(25);  
        var result = sqrt * pow % 2;  
        System.out.print(result);  
    }  
}
```

- A. 18.0-1.0
- B. 9.0-0.0
- C. 18.0-0.0
- D. 9.0-1.0
- E. none of them

29. What is the output of code below if `System.gc()` is executed successfully?

```
public class GCSample {
    public String name;
    public GCSample(String name) {
        this.name = name;
    }
    public static void main(String[] args) throws InterruptedException {
        var a = new GCSample("a");
        var b = new GCSample("b");
        var c = b;
        b = null;
        new GCSample("empty");
        a = null;
        System.gc();
        Thread.sleep(2000);
    }
    @Override
    protected void finalize() throws Throwable {
        System.out.println("destroy : " +this.name);
    }
}
```

- A. code will not compile
- B. destroy : a
- C. destroy : empty destroy : a
- D. destroy : b destroy : a
- E. destroy : b destroy : empty

30. Select statements that are true about garbage collection(gc).

- A. System.gc() will immediately clean eligible objects.
- B. Object that has at least one reference is not eligible for gc.
- C. Primitives and Objects are eligible for gc when they lose all their references.
- D. When Object and their references are out of scope they become eligible for gc.
- E. Immutable class is not eligible for gc.

31. What is the output of code below?

```
public class Main {  
    public static void main(String[] args) {  
        var datetime = LocalDateTime.of(2023, 1, 30, 10, 10);  
        datetime.plus(10, ChronoUnit.YEARS);  
        datetime.plus(2, ChronoUnit.HOURS);  
        System.out.println("year: " + datetime.getYear() + " month: " +  
datetime.getMonth() + " hours: " + datetime.getHour());  
    }  
}
```

- A. code will not compile
- B. year: 2033 month: JANUARY hours: 12
- C. year: 2033 month: FEBRUARY hours: 12
- D. year: 2023 month: JANUARY hours: 10
- E. year: 2023 month: FEBRUARY hours: 10

